Philosophy 415: Question Set #3

Due on Wednesday, Novemeber 6, 2019

University of New Mexico 2:00 PM

Adalida Baca

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1

To prove Gödel's first incompleteness theorem, we shall prove that in any system of a certain sort there is a sentence GS such that the following holds:

1. $GS \iff \sim \text{Prov}(\text{“}GS\text{”})$

(where Prov is a "provability predicate" for that system)

According to the standard interpretation of the theorem, this shows that there is a truth formulable in the language of the system which the system cannot prove. How does it show this and what assumptions must we make in order to interpret the theorem in this way? What sort of system must we be working with in order to build the Gödel sentence GS?

This Gödelian sentence states: There exists a Gödelian Sentence (GS) if and only if it is not provable in "GS". Now let's also say this sentence is used in the formal system of Principia Mathematica (PM). If we do this, we are saying GS is derivable from the PM axioms if and only if "GS" is not provable in PM.

Now, we know right away that this sentence contains self-reference similar to the Liar Paradox since they both reference themselves. However, this sentence differs from the Liar Paradox because the Liar Paradox is saying of itself that it is either true or false, whereas the GS is saying of itself that it can't be provable. Truth and provability are different because the GS may be true, but it just might not be provable.

Okay well, we at least know that GS is either false or true. First, let's assume it's false, then the sentence would say: GS can't be proven in PM is false which would make it true that it is provable in PM. If it were provable in PM, then a proof of GS would exist. So, if PM is both sound and only proves true things, then it itself must be true. However, we started our derivation by assuming it was false, which brings us to a contradiction. Therefore, since it can't be false it must be true.

Now, let's assume it's true. Originally GS said it was not provable and we now know this is true. Therefore, we can derive that GS is true, but not provable. We did not reach a contradiction, just a frustrating situation, since we don't also have a contradiction by assuming it's true. For instance, in the Liar Paradox, no matter if we assume it's false or true, we still arrive at a contradiction; however, that is not the case here.

Therefore, we have derived that there are truths which are not provable in PM or any formal system that has equivalent powers. This aligns with Gödel who argued that PM and similar systems are incomplete. But, how do we know which system we need to build GS in the first place? Sadly, this will return a negative answer that no formal system for mathematics exists that is both sound and complete. So, any system that can model (even as little as) whole-number arithmetic could build the GS statement, but it can't prove it.

2

Using the symbolism we developed in class, please give the machine table for a Turing machine that adds two nonzero finite whole numbers, m and n. You may assume:

i) m and n are each initially entered as a finite string of "1"s (e.g. '7' is represented by a string of seven "1"s) and they are separated by exactly one "0"

ii) aside from m and n themselves, the tape initially contains only "0"s

iii) the 'read/write' head of the machine is started somewhere in the string of "0"s to the left of the beginning of m, and is started in state "A".

1. $\langle A, 0 \implies 0, R, A\rangle$ [If in state A and you find a 0, leave it the same, move right, and stay in state A]
2. $\langle A, 1 \implies 0, R, B\rangle$ [If in state A and you find a 1, change to 0, move right, but change to state B]
3. $\langle B, 1 \implies 1, R, B\rangle$ [If in state B and you find a 1, leave it the same, move right, and stay in state B]
4. $\langle B, 0 \implies 1, end\rangle$ [If in state B and you find a 0, change it to 1, end]

3

In 1925, in laying out his “formalist” program, David Hilbert wrote, “The right to operate with the infinite can be secured only by means of the finite.” What did he mean by this, and how does his thinking about the finite and the infinite underlie Hilbert’s conception of a formal system? What does Gödel’s result about the incompleteness of formal systems for arithmetic show us about the “right to operate with the infinite” that Hilbert speaks of?

Hilbert thought that “The right to operate with the infinite can be secured only by means of the finite”. What does this mean? Well, it means that Hilbert believed that our ability to grasp an understanding of the infinite is possible if and only if we use finite reasoning to justify the claims of the infinite. This would mean that we would need proofs that followed a mechanical procedure which could be carried out in a finite number of steps. We need this to be carried out in a finite number of steps because if not, then no one will live long enough to see the outcome of the procedure.

Moreover, this “mechanical procedure” or “proof” entails that it’s part of a formal system. But how do we distinguish good formal systems from bad ones? Well, at minimum we would need a formal system which holds the property of being consistent. Moreover, we also want this formal system to prove its own consistency. If it could not prove its own consistency, then we would have to use another formal system (or language) to explain the consistency of the original formal system. If this is the case, then we also need a third consistent formal system to explain the second consistent formal system which explains the first formal system. Therefore, to avoid this, we need our formal system to be consistent such that it could prove its own consistency.

Furthermore, Hilbert is really asking two questions. First, if a system which models arithmetic is both complete and sound and second, can any formal system correctly answer all well-defined mathematical questions. Gödel’s result about the incompleteness of formal systems for arithmetic shows us that this cannot be done because any formal system is either complete or sound but not both. We need both in order to have a formal system that can correctly answer all well-defined mathematical questions.

Therefore, since Hilbert argues that to understand the infinite, we have to use finite reasoning to justify the claims of the infinite and Gödel proved this was not possible, then we can never operate with the infinite.

4

In order to write the Gödel sentence in *Principia Mathematica*, it is necessary to find a way for PM sentences to "talk about" or make reference to the formal properties of other PM sentences (or even possibly themselves). With reference to the ideas of recursivity, representibility, and Gödel numbering, explain briefly how Gödel accomplishes this.

Gödel accomplishes this by creating a system that incorporates a kind of self- reference. This system is also referred to as Gödel numbering. We use Gödel numbering because we can effectively talk about systems and their properties of its formulas within the system itself. This is done by code-numbering the symbols and the formulas composed of the symbols. If done correctly, each formula should correspond to exactly one whole natural number. By doing this, we can talk about all the properties a formula has including if it's provable or not. Not only can we do that, but this will allow us to build predicates that contain symbolic strings if and only if it's provable in the system.

Since we can talk about strings, then we can also talk about the property of strings. In doing so, we can closely examine those that are in WFF's and are also theorems (provable from the axiom). This ultimately leads us to demonstrate that these strings are provable by means of the predicate function and we will even be able to pick out the ones with correct Gödel numbering. However, to show this, we have to discuss recursive functions.

A general recursive function is a function that can be defined by finitely many rules. Recursive functions are functions which are computable by Turing machines. This will be helpful in deciding which sets are decidable. For instance, a set is decidable if and only if we have a computable way to decide its membership. This is super helpful because recursive functions can let us identify each set with its characteristic function. If the characteristic function returns a 1 then it's a member and if it returns a zero, then it is not a member. Therefore, we can state that a set is decidable if and only if its characteristic function is recursive.

Lastly, if the function is recursive then it is representable in the system. Representability is important because for each set that is recursive, there exists a statement in the system that holds of only members of the sets. This will let us talk about any number which may also be a Gödel number for a sentence. By using Gödel numbers we are able to make any claim we want about a specific sentence.

5

[Extra Credit]. In 1963, discussing the impact of his 'incompleteness' theorems, Gödel wrote: "Before my results had been obtained it was conjectured that any precisely formulated mathematical yes or no question can be decided by the mechanical rules of logical inference on the basis of a few mathematical axioms. In 1931 I proved that this is not so. i.e.: No matter what and how many axioms are chosen there always exist number theoretical yes or no questions which cannot be decided from these axioms. Combining the proof of this result with Turing's theory of computing machines one arrives at the following conclusion: Either there exist infinitely many number theoretical questions which the human mind is unable to answer or the human mind . . . contains an element totally different from a finite combinatorial mechanism [such as a Turing machine]. . . I hope I shall be able to prove on mathematical, philosophical and psychological grounds that the second alternative . . . holds" How might we decide between these two "alternatives," and based on what we've learned about Gödel's results so far, do you agree or disagree with Gödel's own conclusion? Why?

I would say that the first "alternative" suggests a limitation of the human mind whereas the second "alternative" suggests that the human mind has something special about it that distinguishes it from a machine. I would say I am on spectrum of both alternatives.

For instance, I believe humans are capable of doing great things; however, we are only human. We can and do tell these great big machines to compute the things we sometimes can't but want. I don't see anything wrong with that, honestly. It saves me and tons of other people a lot of work! In school I would always forget to add the negative to something or it would take me a while to add 12 different numbers together. So, I see the machine and the first "alternative" as an easy way to compute the things I want while making it easier on myself.

However, with that, the human mind has the power or special "element" that the machine can never have, which is what makes us capable of even more than the machine. For instance, someone has to tell the machine to do the calculations or perform x action because they don't have the "element" to do it on their own (varies with machine learning capabilities). I find that a lot easier and more appealing to understand and do while still getting the result I want without having to do what the computer is doing.

So, to answer the question of either the first or second "alternative", my answer is yes. I don't think this situation has to be an exclusive or. I think it's an inclusive or, which is why I put myself on the spectrum between the two.